ZMI

UltrasoundTherapeutic Device

MODEL: ULS-1000

User's Manual

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Notes in accordance with EC directive and Medical Device Directive (MDD)

The device is in accordance with the EC regulations for medical devices directive (93/42/EEC) as amended by (2007/47/EC) and therefore carries the CE sign with the registration number of the notified body for medical devices.

According to the MDD, **ULS-1000** is a **class IIa** device.

The manufacturer is only responsible for the safety, operational reliability and functionality of the device if:

- The device is used in accordance with the instructions for use
- The electrical installation of the location where the device will be used meets the respective current requirements of electrical safety
- The device is not used in hazardous environments and humid locations
- Mountings, amplifications, re-adjustments, modifications or repair works are carried out only by personnel authorized by the manufacturer
- The operator regulation of this EC directive is observed within the scope of MDD.

Technical support may be obtained by the manufacturer, dealers or service authorized by the manufacturer.

ULS-1000 is an electronic device. For its disposal the according regulations for electronic devices have to be observed.

On request, the manufacturer will provide you with further technical descriptions for all repairable parts of the device, such as circuit diagrams, spare parts lists, and adjustment instructions as far as these are necessary for the qualified technical staff of the operator.

Comments on electromagnetic compatibility (EMC)

Medical, electrical devices are subject to special precautions concerning the EMC. They must be installed and operated according to the EMC-advice given in the accompanying documents. In particular medical, electrical devices may be influenced by portable and mobile RF-communication devices.

This device, **ULS-1000**, conforms to EMC measurement standard, **EN60601-1-2**.

The manufacturer guarantees the conformity of the unit with the EMC-requirements only when using accessories. The usage of other accessories may cause an increased emission of electromagnetic disturbances or may lead to a reduced electromagnetic immunity.

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V3.0 20140116

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1. Function Overview

ULS-1000 Ultrasound is a unique physical modality in that it uses high-frequency sound energy to produce both thermal and non-thermal effects. On tissue it has more clear and prompt thermal effect than any of the electromagnetic heating modalities.

Product Features:

1. Controlled Depth of Penetration
   The ULS-1000 enables you to treat soft tissue injuries, superficially with 3MHz and instantly at the push of a button, treat deeper injuries with 1MHz - without switching transducers.
2. Two transducer sizes for selection.
3. Microprocessor controlled digital screen
4. Easy-to-read digital screen offers accurate indications of all treatment parameters, automatic self-test, ERA and BNR.
5. Excellent BNR and ERA
   Our superior beam non-uniformity ratio (BNR) and effective radiating area (ERA) makes for highly uniform, thorough applications
6. Over heat monitoring and skin contact protection
   The device is enabled to decrease the intensity automatically when the transducer head is overheated. Meanwhile, the function of skin contact provides a further protection for monitoring the ultrasound output. Once the transducer doesn't contact the skin, the output is automatically shut down to avoid the burning of transducer.
7. 20 set memories for saving the modality settings.

Purpose:

To relieve pain, heal wounds, provide micro-massage, promote blood effusion and permit phonophoresis.

Modalities:

Transducer	Mode	Duty cycle	Selective output	Selective therapy time
Medium	1 MHz mode	continuous 1:1 1:3 1:9	0~6 watts	0-30 minutes
	3 MHz mode		0~6 watts	
Small	1 MHz mode	continuous 1:1 1:3 1:9	0~3 watts	0-30 minutes
	3 MHz mode		0~3 watts	

※ Adjustable power output: Medium Transducer 0.2 watt every graduation
Small Transducer 0.1 watt every graduation

※ Intensity (w/cm^2) = Output power (watts) / ERA (cm^2)

※ Adjustable time ±1 minute every graduation.

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2. Precautions and Warnings

2.1 Be cautious of the following items

	If device function or operation is abnormal, please contact your vendor for repair service.
	Gradually increase the output intensity / power to patient tolerance while monitoring the LCD screen on the ULS-1000.
	Therapists must understand the symptoms and prognosis prior to use of device to determine appropriateness of use.
	Therapists must clearly communicate to user to inform the therapist to power off device if user is in a state of discomfort.
	Review the indication, contraindication, warning, and precautions of the ULS-1000 unit before beginning combination treatment with the ULS-1000.
	Decrease the output intensity / power before varying the duty factor or frequency. At the termination of treatment, decrease the output intensity / power to zero and remove the transducer head from the patient before turning the ULS-1000 off.
	Caution should be taken with patients exhibiting psychological or physical hypersensitivity to therapeutic ultrasound. Several attempts should be made to place them at ease so that their confidence and cooperation can be gained during the treatment.
	This product is to be used by professionally trained physicians or therapists only.
	Avoid use of product in unclean environments.
	During use, make certain to maintain good product ventilation of heat. Avoid use of device on soft or uneven surface that may potentially provide blockage of ventilation.
	When not in use, place device in Standby mode to save power.
	Avoid contact with water or damp environments.
	The ULS-1000 must use its designate probes identified by the probe ID, because the power has been calibrated during manufacturing.
	Power off and remove power cable before cleaning device.
	Do not immerse any part of the operator's body into the water bath during underwater techniques since the long-term biophysical effects of ultrasonic energy exposure have not been evaluated.
	Do not apply therapeutic ultrasound directly over the heart, eyes, spinal cord or on the carotid sinuses, cervical stiletto ganglion or vagus nerve.
	Do not store in direct sunlight.
	Do not operate the ULS-1000 near flammable or explosive gases.

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	Do not connect the ULS-1000 to any therapeutic or non-therapeutic line-powered equipment unless the equipment's line current is completely isolated from the ULS-1000 and the patient.
	Do not connect the ULS-1000 to any wall outlet that has not been properly grounded.
	Do not use any extension cords or two-prong or three-prong adapters between the line cord plug and the wall outlet.
	Do not share wall outlets with any high frequency devices, e.g., short wave or micro-wave diathermy as well as with any motorized equipment, e.g., whirlpools or traction machines.
	Do not operate the ULS-1000 with line fuses that are not of the correct current rating listed on the rear nameplate.
	A potential electric shock hazard exists once the ULS-1000's outer casing has been in part or fully removed. Service and repairs should be performed only by qualified service personnel.
	The electrical output of the ULS-1000 may possibly be hazardous. This unit should be used by and / or under the direct supervision of a qualified medical practitioner.
	Please avoid operations in extreme environments (below 0 °C or 30 °F or above 50 °C or 122 °F). Extreme environments may affect normal function.
	Please avoid using device near or on other heat generating devices such as microwaves and heaters.
	Please avoid using device near or on other steam generating devices such as steamers or water heaters.
	Please avoid using device near high voltage or other high radiation emitting device.
	Please use device in a dry environment. Water, steam, and other liquids will have significant deteriorating effects on electronic components.
	When using an extension cord with this device, do not use the same extension cord with other devices simultaneously to prevent surges, shortages and possible electrical caused fires.
	Please do not open the device. The company reserves the right to repair and service only untampered devices. The company will maintain the right to refuse service for any tampered devices.

2.2 Do not use under the following conditions

	Users with low immune systems or auto-immune conditions
	User with heart conditions
	User is pregnant
	User with abnormal blood disorders
	Users with third degree burns

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	Users with transmittable diseases
	Users with brain disorders
	Users with irregular heartbeats
	Users with pace makers and other artificial devices
	Users with invasively implanted synthetic life sustaining devices
	Users with malignant tumors or cancerous lesions

2.3 Usage requiring physician direction

	Users with emergency external wounds, bone fractures, and bleeding
	Users with severe loss of sensation of skin
	Women during menstrual cycle

2.4 Safety Criteria

- Always connect the equipment to a grounded electrical socket.
- The precision electrical and electronic components should never be taken apart, dropped, or be subjected to sudden impacts.
- Be careful to avoid subjecting the equipment to prolonged periods of high temperature, high humidity, or dusty conditions.
- Be sure that the room in which the equipment is being used is free from flammable gases or vapours.
- Turn-off the equipment prior to disinfecting the work area.
- Do not use volatile liquids (such as thinners) to clean the machine. Use a soft cloth, or one that has been dampened with a neutral detergent solution.
- Avoid operating the equipment in close proximity to a microwave appliance as output could be affected.
- Do not use this equipment in conjunction with a high frequency surgical appliance, as burns could result.
- Do not allow the equipment to be used by anyone other than a qualified operator and, then, only for the purposes of Electro-therapy.
- Do not drop the transducers as this could damage the crystals.
- Do not switch the probe randomly. The ULS-1000 must use its designate probes identified by the probe ID, because the power has been calibrated during manufacturing.

NOTE: The liquid crystal display may not function, or require more time to function, at low temperatures.

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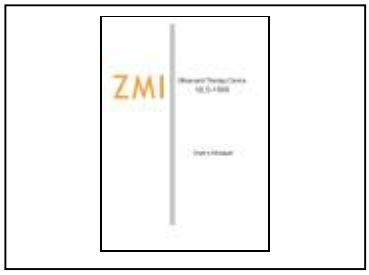
3. Standard Accessories

The complete unit includes the following standard accessories:

Item no.	Quantity	Description
ULS-1000	1	Ultrasound therapy device
TUM01000A	1	Probe medium transducer (5 cm²)
TUS01000A	1	Probe small transducer (2 cm²)
OPW0060*	1	AC cord
KMN0240*	1	ULS-1000 Operation manual

- ULS-1000 x 1

- Manual x 1



- AC cord x 1

- Medium Transducer x 1

- Small Transducer x 1

Note: Standard accessories come with one piece each of Medium and Small Transducer. Assortments of Medium and Small Transducer size(s) are available upon request.

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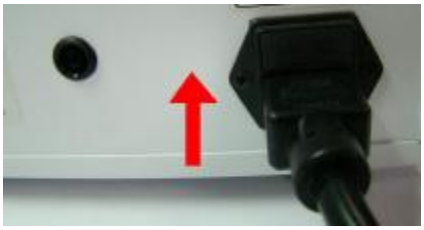
4. Installation

Power Cord and Probe Installations

a. Plug the AC power cord into the device and connect to the power supply 230 V with 50/60 Hz



b. Select the desired transducer and plug in the PROBE1 socket or PROBE2 socket.

c. Clockwise screw the nut on the socket

d. Switch on the power and start to operate the ULS-1000

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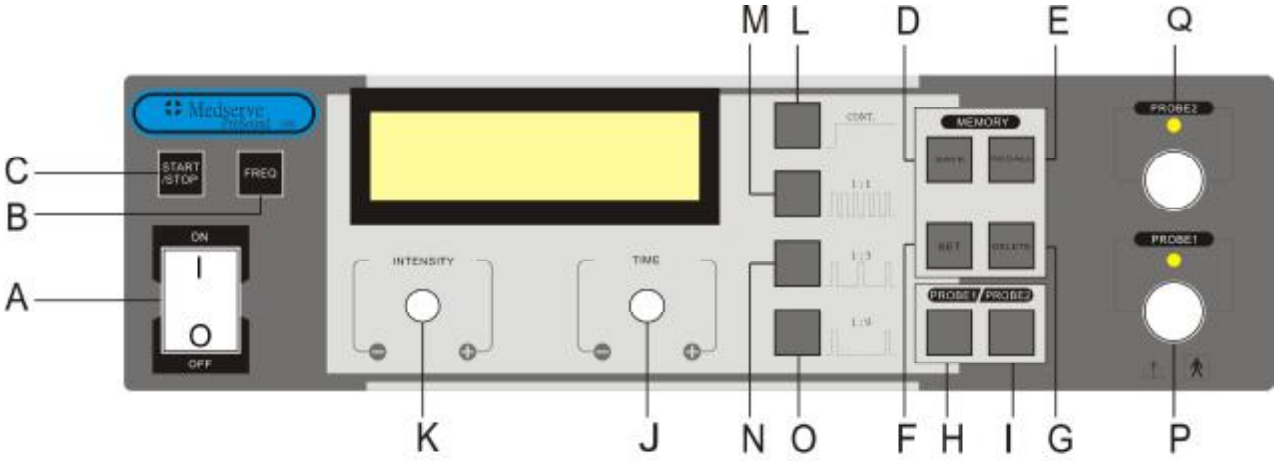
5. Controls and Functions

5.2.1 Front View



A. POWER ON/OFF SWITCH

- Main power switch. Press up for ON, Press down for OFF.

B. FREQUENCY SELECTION 1 MHz/3 MHz

- Switch to 1 MHz or 3 MHz for required therapy

C. START/STOP SWITCH

- Start or stop the treatment

D. SAVE

- Save the settings of frequency, intensity, time, mode, and probe selection.

E. RECALL

- Recall parameters from saving memory.

F. SET

- Set parameters of frequency, intensity, time, mode, probe selection, and control the function of skin contact.

G. DELETE

- Re-call parameters in Memory 2

H. USE PROBE 1 THERAPY

- Press this button to use Probe 1 therapy. If transducer not connected, LED will not illuminate and unit switches automatically to Probe 2.
- Press PROBE1 and turn on the main power switch simultaneously to get PROBE1 message as below: (this function is for maintenance reference only)

I. USE PROBE2 THERAPY

- Press this button to use Probe 2 therapy. If transducer not connected, LED will not illuminate and unit switches automatically to Probe 1.
- Press PROBE2 and turn on the main power switch simultaneously to get PROBE2 message (same above PROBE1, for maintenance reference only).

J. TREATMENT TIME SELECTION (0~30 MINUTES)

- Turn knob to the right to increase therapy time, turn knob to the left to decrease therapy time

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K. INTENSITY SELECTION

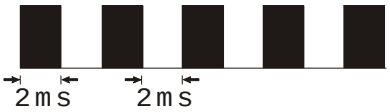
- Turn knob to the right to increase intensity, turn knob to the left to decrease intensity
- Adjustable intensity: Medium Transducer 0.2 watt every graduation
  Small Transducer 0.1 watt every graduation

L. CONT.

- Full frequency therapy

M. 1:1

- Pulse rate 1:1 mode



N. 1:3

- Pulse rate 1:3 mode

O. 1:9

- Pulse rate 1:9 mode

P. PROBE 1 SOCKET

- Plug in a transducer

Q. PROBE 2 SOCKET

- Plug in a transducer

R. PROBE 1 LED INDICATOR

- The LED indicator lights when PROBE 1 outputs; it turns off when output stops.

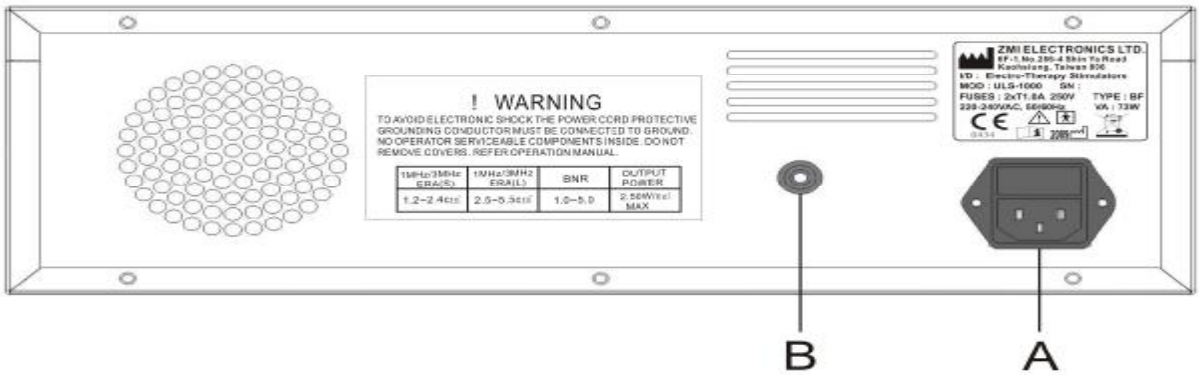
S. PROBE 2 LED INDICATOR

- The LED indicator lights when PROBE 2 outputs; it turns off when output stops.

T. LCDM DISPLAY

- Displays treatment parameters.

5.2.2 Rear View



A. POWER CORD SOCKET

B. STIM CONNECTOR

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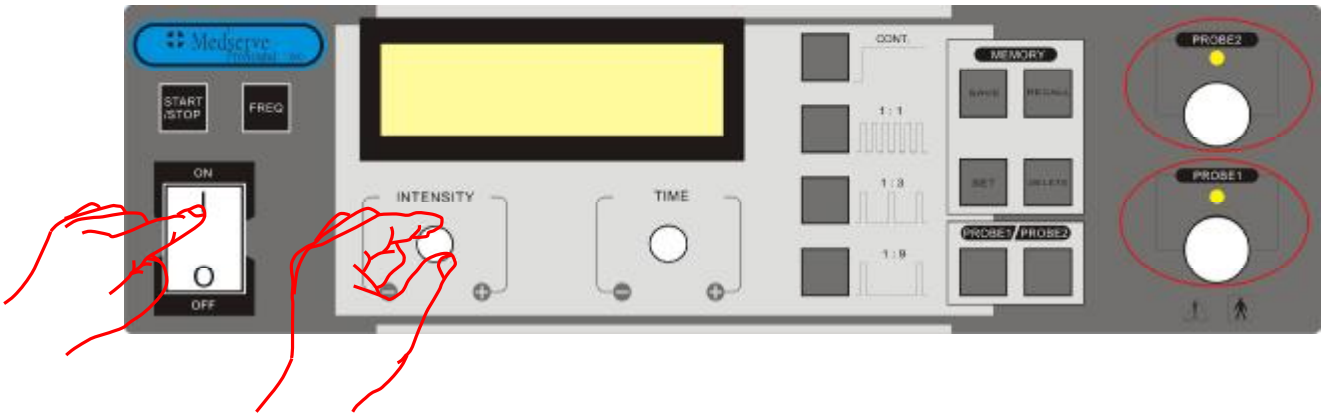
6. Operation Instructions

6.1 Parameter Setting

Step1: Plug the medium and small transducer heads into the sockets, PROBE 1 and PROBE 2, as placements of circle.



Step 2: Switch ON the power and the display will show below

Mode: 1 MHz - 1:9	Probe1
Power: 0.0 W	ERA: 3.7 cm^2
Intensity: 0.00 W/cm^2	
Stop	Min 00:00

Step 3: Refer to the Appendix to set the desired parameters and modalities.

Step 4: Press the button PROBE1 or PROBE2 to select the target probe.

Step 5: Press the button FREQ. to select the desired treatment frequency 1MHz or 3 MHz.

Step 6: Rotate the knob INTENSITY to set the treatment intensity from range 0-6 watts.

Step 7: Rotate the knob TIME to adjust the time range 0-30 minutes.

Step 8: Select desired continuous mode CONT., or intermittent mode 1:1, 1:3, 1:9.

Step 9: Press the button SET to turn on/off the function of Skin Contact Detection.

Mode: 1 MHz - 1:9		Probe1
Power: 0.0 W		ERA: 3.7 cm^2
Intensity: 0.00 W/cm^2		
Stop	Skin On	Min 00:00

Mode: 1 MHz - 1:9		Probe1
Power: 0.0 W		ERA: 3.7 cm^2
Intensity: 0.00 W/cm^2		
Stop	Skin Off	Min 00:00

Step 10: Press the button START/STOP to start treatment. The outputting probe indicator lights. Once the timer counts down to zero, the treatment is over, at the same time the buzzer is alarmed and the output is shut down.

Mode: 3 MHz – Cont.	Probe1
Power: 6.0 W	ERA: 4.3 cm^2
Intensity: 1.39 W/cm^2	
Start	Min 28:34

Mode: 3 MHz – Cont.	Probe1
Power: 6.0 W	ERA: 4.3 cm^2
Intensity: 1.39 W/cm^2	
Stop	Min 30:00

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6.2 Parameters SAVE, RECALL and DELETE

6.2.1 SAVE

a. During the Non-treatment status, press the button SAVE to enter into the save mode.

b. Edit the name of memory by using knobs INTENSITY and TIME. And then press the button SET to save parameters into the memory. As long as either parameters or name of memory is set as Zero, the system will alarm three beeps to represent the failure of saving. If 20 memory sets are over full saving, the system will show the message below.

```
The memory is full.
Please delete some
memory before saving.
Press FREQ key to exit.
```

c. If not doing any action, press the button FREQ. to exit and return to the initial screen.

6.2.2 RECALL

a. During the Non-treatment status, press the button RECALL to enter into the recall mode.

b. Adjust the knob INTENSITY to select the desired memory set. Press the button SET to recall the memory set.

c. If there is no memory for recall, system will show the message below.

```
The memory is empty.
Press FREQ key to exit.
```

d. If not doing any action, press the button FREQ. to exit and return to the initial screen.

6.2.3 DELETE

a. During the Non-treatment status, press the button DELETE to enter into the delete mode.

b. Adjust the knob INTENSITY to select the desired memory set. Press the button SET to delete the memory set.

c. If there is no memory for delete, system will show the message below.

```
The memory is empty.
Press FREQ key to exit.
```

d. If not doing any action, press the button FREQ. to exit and return to the initial screen.

6.3 Function of Skin Contact Detection

- Press the button SET only can turn on/off the function of skin contact detection. Once the function is set, the on/off status will be kept.
- Turning on the function of skin contact detection, once the transducer does not contact the skin for 2-3 sec, the ultrasound output will be automatically shut down, timer will be paused, the LED on the membrane will glitter, and the buzzer will alarm at the same time until the

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transducer contact the skin again for at least 0.5 sec, system will recover the output.

Note: This function is unable to work with electrical stimulator for simultaneous treatment.

6.4 Combination with Electrical Stimulator

- ULS-1000 is able to combine with the electrical stimulator for simultaneous treatment. The connection is to plug the POSITIVE output of stimulator into the jack as shown the picture below. At the same time, the NEGTIVE output (electrode) is placed on the part of treatment body. And then the therapist can apply the ultrasound probe to be both ultrasound and electrical stimulation outputs.

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7. Troubleshooting

7.1 Over Heat Protection

If the transducer sensor detects the temperature is over 41 °C, the phrase “Over heat” will flash on the display and the unit will emit three beeps. The unit will stop the output until the temperature is down to the proper level. And then the unit will beep twice to inform that the device has recovered. The therapist can press the button START/STOP to continue the unfinished treatment or reset the parameters and then press the button START/STOP to restart a new treatment.

```
Mode: 3 MHz – Cont.        Probe1
Power: 6.0 W          ERA: 4.3 cm2
Intensity:  1.39 W/cm2
Over heat                 Min 28:24
```

Note: Make sure that output is switched-off when the transducer is not in use. Temperature will rise rapidly if the output is switched-on and the transducer is not in contact with the skin.

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7.2 Error Message

When the power is switched on, the unit will automatically detect that the connection of probes and the inner memory of probes are OK or not. If not, the unit will alarm and display the error message. Please refer to the below Error Messages:

Error Message 01 Probe not connected !

Error Message 02 NVRAM read/write error !

Error Message 03 Probe not set-up !

Error Message 04 Probe not calibrated !

Error Message 05 ERA & BNR not in probe !

Error Message 06 Wrong information in NVRAM !

Error Message 07 Different probe version !

Error Message 08 Thermistor cable broken !

Once the Error Message is happening, please switch-off the power and reinstall the ultrasound probe. And then switch on the power again, if the Error Message is still existing, it means the ultrasound probe is broken. Please contact the distributor for repair.

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Note: No matter in any kind of circumstances, please Do Not screw out the transducer head in case of break.

8. Maintenance

Functionality, reliability and safety characteristics of ultrasound therapy devices are only guaranteed if properly used in accordance with the operating instructions. Safety control, maintenance, repair and modifications shall be carried out by the manufacturer or the service agents authorized by him. In case of a failure, parts which influence the safety of the device shall be replaced only by original spare parts of the manufacturer.

The device does not contain any parts which need maintenance work done by the user.

8.1 Safety controls

The device is subject to the provisions of the MDD. The safety controls have to be carried out on the basis of this directive. Thereby, the operator regulation has to be especially observed.

Irrespective of the legal rules or beyond the scope of the MDD, it is recommended to have the device checked by the manufacturer or by an authorized service agency at 12-months intervals.

The check shall consist of at least the following:
• Electrical safety check in accordance with the test plan of the manufacturer
• Check of the device in respect of external integrity
• Check of all display and operating elements in respect of damage
• Check of all inscriptions in respect of legibility

8.2 Cleaning, disinfection and care

For cleaning and disinfection of the device and its accessories there should not be used any agents containing higher portions of phenol derivates, alcohol, compounds of chlorine or peracetic acid. It is recommended to use disinfectants on aldehyde basis. The device is not suited for heat sterilization or for sterilization with gases.

Attention
Before cleaning or disinfection unplug the mains plug out of the socket!
The device and the ultrasound probes are suited for wiping disinfection. It has to be observed that no liquids enter the device or the ultrasound probes. Never shall the plug or socket get wet.

8.3 Disposition of Waste

According to European Union Directive on Waste from Electrical and Electronic Equipment (WEEE 2002-96-EC), this device, ULS-1000, follows the national requirement to dispose unit.

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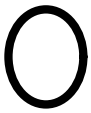
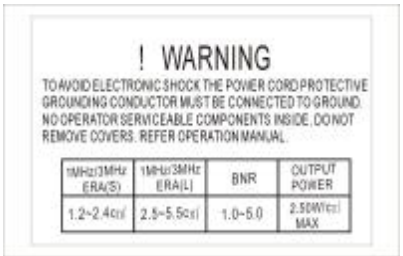
9. Mark Description

	Attention, consult ACCOMPANYING DOCUMENTS (IEC Publication:348)
	Type BF Equipment (IEC Publication:417-5008)
	Power Off (IEC Publication:417-5008)
	Power On (IEC Publication:417-5007)
	Medical Device Certificate by DNV notifying body
	Consult instructions for use (ISO 7000-1641)
	The label content varies with real manufacturing year
	According to European Union Directive on Waste from Electrical and Electronic Equipment (WEEE 2002-96-EC)
	Label for warning & output signal specification
	Manufacturing label and power supply specification.

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10. ZMI Limited Warranty

Warranty Period

This ZMI product is warranted against defects in material and workmanship for a period of two (2) years from the date of original purchase. This warranty extends to any owner of this product during the warranty period. For service under this warranty, the product must be returned to the ZMI service department or ZMI designated service partner within the warranty period and within thirty (30) days after discovery of and notification to ZMI of the defect.

Limited Warranty

ZMI will, at its discretion, repair or replace without charge any part(s) which prove to be defective during the warranty period provided that the repairs or replacement part(s) are serviced by ZMI.

Warranty Exclusion

The warranty is void if any or all of the following have occurred:

(1) The product contains repairs or replacement parts not furnished by ZMI.
(2) The product contains damage resulting from misuse, accident or negligence.
(3) The product has been tampered with and/or altered, including serial number removal, alteration, or effacement.

Shipping

Shipping charges for the transportation of the defective product to and from ZMI will be covered by ZMI for a period of ninety (90) days from the date of original purchase. After ninety (90) days, and for the remainder of the warranty period, ZMI will pay for the return postage only.

IMPORTANT NOTE

DO NOT SHIP THE PRODUCT TO ZMI WITHOUT FIRST SECURING AUTHORIZATION TO DO SO. EQUIPMENT SENT IN WITHOUT FACTORY AUTHORIZATION WILL NOT BE ACCEPTED.

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11. GLOSSARY

BNR	Beam Non-uniformity Ratio
ERA	Effective Radiating Area
Ultrasound	Sound waves that vibrate at frequencies, beyond the hearing power of human beings (above 16,000 Hz). Commercial and military applications include ultrasonic cleaning, gauging, cut-ting, detection instruments, and welding.
Ultrasound Frequency	Also called an ultra-audible frequency. 1. Any frequency above the audio ranges, but commonly applied to elastic waves propagated in gases, liquids, or solids. 2. Sound frequencies which are above the range of human hearing; approximately 20,000 Hz and higher.
Ultrasound Transducer	A device which takes the electrical oscillations produced by the ultrasound generators and transforms them into mechanical oscillations. Typical transducer materials are piezo-electric (e.g., quartz or barium titanate) or magnetostrictive (e.g., nickel).
NVRAM	Non-volatile Random Access Memory

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12. Technical Specifications

Power Supply	AC 230 V 50/60 Hz
Power Consumption	73 W
Ultrasound Frequency	1 MHz/ 3 MHz
LCM	4 line X 24 character
Operation Temperature/Humidity	0~40 °C / 90%RH below
Storage Temperature/Humidity	-20~60 °C / 90%RH below
Dimension	W342 mm X D270 mm X H127 mm
Net Weight	7.2 kg

Ultrasound Generator

Frequency	1 MHz ± 10% or 3 MHz ± 6.7%
Pulsed Mode	Output is continuous or pulsed. In pulse mode output is 100% square wave modulated. Power level is adjusted with intensity controls pulse amplitude.
Continuous Mode	Output is on as long as the timer has time remaining and the output control is turned up.

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Appendix

ULTRASOUND DOSAGE TABLES (RECOMMENDED*)

Condition	Area of Treatment	W/cm^2	Minutes	Treatments	Remarks
Abscess	Local or peripheral	0.5-2	2-8	5-10	
Anal fissure	Local and perifocal water	1-2.5	4-6	12-15	Under
Arthrosis	Paravertebral and apophysis	1-3	5-15	10-20	
Large articulations	Local and periarticular	2-3	5-15	10-20	
Small articulations	Local and periarticular	1-2	2-5	8-12	
Bechterew's disease	Paravertebral, apophysis	1-3	5-15	15-30	
Bronchitis mbronchiectasis	C-3,D1-12b	1-3	5-15	10-25	
Bronchial asthma	Paravertebral C3-D1-12 stellate ganglion from front, paracardiac	0.5-2	5-10	8-15	Frequents essions, small dosage
Bursitis	Local and peripheral minimum dosage.	0.6-1.5	5-10	10-18	In acute conditions
Causalgia	Local, periarterial and Perineural	0.5-2	5-10	10-25	
Cellulitis	Local	1-2	5-10	5-15	
Cholecystitis	Local, paravertebral D-7-10	1-2.5	5-10	10-20	
Claudication intermittent	Tibial and along thigh, L4-5,S1-2	0.5-2	5-15	5-20	Avoid excessive dosage
Coccygodynia	L4-S3 paravertebral and local	1-2	5-10	5-12	
Coxitis	Local on every side	1-3	5-12	10-15	
Dupuytren's contracture	Local, along cervical column, paravertebral C5-7	1.5-3	10-15	10-20	
Eczema	Local, periarterial	1-1.5	5-10	5-10	Under water
Endarteritis obliterans	Periarterial	1-2	5-10	5-15	
Epicondylalgia	Local, cervical column, C-4	1-3	5-8	5-10	
Erythema nodusum	Local	0.8-2.5	5-10	10-15	
Erythromelalgia	Fem. art.	1-2	5-10	10-25	Side only heavier dosage
Fistula	Local	1-2	5-10	10-20	
Furuncle anthrax	Local,perifocal	0.7-1.5	5-8	8-12	
Haemorrhoid	Local and perifocal	0.8-2	5-6	10-15	Under water
Herpes zoster	Paravertebral and local	2-3	5-10	5-10	
Hypertension, essential	Paravertebral C4-D9 treatments	1-2	5-10	5-15	Stop if negative afterfive
Lumbago	L3-S1 paravertebral and apophysis L2-S4	1-3	10-15	2-10	Daily
Lymphadenitis	Local	0.5-2	5-10	5-15	
Mastitis, puerperal	Local	0.8-2	10-15	10-20	
Myalgia paravertebral segments	Local, corresponding	2-3	5-10	3-20	
Myelitis	At level of medullar focus	0.8-2	5-10	5-10	
Myositis	Local and paravertebral	1-2	5-10	10-20	
Neuralgia intercostal	Intercostal soaces and paravertebrally	0.6-2	5-10	5-15	Beware angina pectoris
Neuralgia neuritis	Local and paravertebral segments	1-3	5-10	5-20	

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Condition	Area of Treatment	W/cm^2	Minutes	Treatments	Remarks
Osteitis	Local and perifocal	1-2	3-10	10-30	
Osteomyelitis	Local	1-2	5-10	10-30	
Otosclerosis	Mastoid apophysis	0.3-1.5	5-8	5-15	Shave
Parotitis	Local	0.5-2	5-10	10-15	
Periarthritis Humeroscapular	Local and inferior cervical Column, stellate ganglion	0.5-3	5-15	10-20	Heavy dosage for Local treatment
Periostitis	Local and perifocal	0.8-2.5	3-10	10-30	
Polyarthritis, chronic	Local and paravertebral erve roots	1-2	5-10	5-15	
Postoperative pai	Local and affected nerve roots	1-2	5-10	4-10	
Prostatitis	Local	1-3	5-10	10-20	
Pruritis	Local	1-2	5-15	5-30	
Pulpitis	Local	0.5-1	3-5	2-4	Through the cheek
Radiculitis	Paravertebral and all along nerve	1-2	5-10	5-15	
Raynaud's disease	Local	1-2	3-8	5-12	
Rheumatism	Local and paravertebral segments	0.8-3	5-15	10-30	
Sciatica	L3-S1 paravertebral, then L5-S1	1-2	5-15	5-20	
Scleroderma	Local and related paravertebral segments	1-2	5-15	10-20	
Sclerosis miliary	Paravertebral	2-3	5-15	10-20	
Sinusitis, frontal	Local	0.3-0.8	4-8	5-10	Through the cheek
Sinusitis, maxillary	Local	1-2	5-10	5-15	
Spondylitis, deformans	Local	2-3	5-10	5-20	
Stumps	Local	2-3	5-15	5-15	
Sudeck's atrophy	Local	1-3	5-10	5-15	Under water
Sudoriparous abscess	Local	1-2	3-5	5-10	Under water
Syndrome scalenus	Local and D4-7	0.6-1.5	5-10	5-10	
Tenovaginitis	Local	1-2	5-10	5-20	
Thomboangiitis, obliterans	Local, periarterial and paravertebral along nerve roots	1-2	2-5	20-40	Twenty daily sessions, interval of 30 days, and 20 more treatments
Thrombophlebits	Local	0.5-2	4-10	5-15	
Thyrotoxicosis	Local	1-2	5-10	5-15	
Tonsilitis,chronic	Maxillar angle	1-2	5-10	10-20	
Tumors	Local	2-3	5-15	10-20	
Ulcus cruris	Local, periarterial paravertebral	1-3	4-15	5-10	Under water
Ulcus ventriculi	D9-10 front, paravertebral D6-10	1-3	5-10	10-20	

* Based on a composite of European and American reports of pathologies and modalities for ultrasound treatment – courtesy of the Bircher Corporation.

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